



**Verified Carbon
Standard**

AÇAÍ REDD+ PROJECT



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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Açaí REDD+ project aims to promote forest conservation and reduce potential greenhouse gas (GHG) emissions from unplanned deforestation and forest degradation. The property where the project is located covers a continuous area of 16,229.09 ha of Amazon Forest, of which 15,715.82 ha are destined for Avoiding Unplanned Deforestation (AUD).

The Açaí REDD+ project is located in the municipality of Envira, in the state of Amazonas (39,90% of the project area), and in the municipality of Feijó, in the state of Acre (60,10% of the project area). Envira is predominantly covered by natural forest formations, accounting for 94.75% of its area, while pasture areas make up 3.61%, with other land cover classes comprising the remainder. Similarly, in Feijó, approximately 93.4% of the territory is characterized by forest formations, while 6.08% is designated for pastures. The site of the Açaí REDD+ project also exhibits a high proportion of forest formations, constituting 99.93% of the property, with only a small portion dedicated to pastures and floodable forests.

According to MapBiomas datas¹, the states of Amazonas and Acre have been identified as having one of the highest deforestation rates per area during the period from 2019 to 2022. Specifically, the municipality of Feijó, ranked twenty-fourth in the ranking of municipalities most affected by deforestation during this period. The accumulated deforestation in this municipality reached the mark of 17.194,1 hectares in 2022, where agriculture is considered the main vector of vegetation suppression².

The project will conserve 15,715.82 hectares of Amazon rainforest and its biodiversity through the surveillance of the project area by land and river performed by outsourced people and companies, monitoring of forest fires³, annual monitoring of forest cover through forest inventories and satellite images, and periodic monitoring of endangered species.

At the end of the project, with the conservation of the total project area, it is estimated a total avoided emissions of 9,922,006.23 tons of Greenhouse Gases (GHG) (tCO₂e) (248,050.16 tCO₂e/year), contributing to the mitigation of climate change.

¹ Mapbiomas. Annual Deforestation Report - RAD 2022. Available at: https://storage.googleapis.com/alerta-public/dashboard/rad/2022/RAD_2022.pdf. Accessed on: 25/03/2024.

² Mapbiomas. Annual Deforestation Report - RAD 2022. Available at: https://storage.googleapis.com/alerta-public/dashboard/rad/2022/RAD_2022.pdf. Accessed on: 25/03/2024.

³ INPE - BDQueimadas. Available at: <http://terrabrasilis.dpi.inpe.br/queimadas/bdqueimadas/>. Accessed on: 09/10/2023.

1.2 Audit History

The Açaí REDD+ project has no audit history.

Table 1.1. Açaí REDD+ project audit history.

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/ verification	N/A			

1.3 Sectoral Scope and Project Type

Table 1.2. Sectoral Scope and Type of the Açaí REDD+ project.

Sectoral scope	14: Agriculture, forestry, and other land use
AFOLU project category	REDD (Reducing Emissions from Deforestation and Forest Degradation)
Project activity type	AUD (Avoided Unplanned Deforestation)

1.4 Project Eligibility

1.4.1 General eligibility

The Açaí REDD+ project is eligible to participate in the VCS program and is aligned with the requirements of Scope 14 of the VCS, the AFOLU (Agriculture, Forestry, and Other Land Use) project category, meeting all related requirements. Scope 14 of the VCS addresses activities aimed at reducing greenhouse gas emissions related to deforestation and forest degradation. The Açaí REDD+ project fits appropriately within this scope as it aims to reduce greenhouse gas emissions through the conservation and sustainable management of forests.

The project encompasses the Unplanned Deforestation (AUD) activity. The AUD area of the project was allocated within the Legal Reserve (RL) and Permanent Preservation Area (APP) zones of the property, as it involves native vegetation that is not subject to legal authorization for deforestation, in compliance with the eligibility of REDD activities as described in the VCS Standard v4.6. These activities are supported by the Verra Methodology for REDD Projects; therefore, this project is eligible under the scope of the VCS Program.

1.4.2 AFOLU project eligibility

The Açaí REDD+ project activities involve the protection of forest areas threatened by deforestation and degradation, qualified as forest for a minimum period of 10 years before the project's start date, with no clearing of existing natural ecosystems during this period. For the eligibility study of areas for the Açaí REDD+ project, the following data were used:

- Mapbiomas data⁴.
- PRODES data⁵.
- INPE data (BDQueimadas)⁶.
- Hydrography and Permanent Preservation Areas (APP) from the Brazilian Foundation for Sustainable Development (FBDS)⁷.
- Satellite images: SPOT-5, Landsat 5, Landsat 8, Sentinel 2 and CBERS.
- Vegetation classified by IBGE (2021)⁸ at a 1:250,000 scale.

These image files were acquired from official data sources, analyzed, and classified to ensure compliance with the VCS requirements outlined in the VM0007 methodology v.1.7 and VMD0007 BL-UP v3.3. These analyses were conducted to verify that the project area consists solely of forested areas for a minimum of ten years preceding the project initiation, covering the period from 2012 to 2022 (Figure 1.1)

⁴ MapBiomas. Available at: <<https://brasil.mapbiomas.org/colecoes-mapbiomas/>> Accessed on: 26/03/2024.

⁵ PRODES. Available at: <<http://terrabrasilis.dpi.inpe.br/>> Accessed on: 26/03/2024.

⁶ INPE - BDQueimadas. Available at: <<http://terrabrasilis.dpi.inpe.br/queimadas/bdqueimadas/>>. Accessed on: 26/03/2024.

⁷ FBDS. Available at: <<https://www.fbds.org.br/>> Accessed on: 26/03/2024.

⁸ IBGE. 2021. Available at: <<https://www.ibge.gov.br/en/geosciences/environmental-information/vegetation.html>> Accessed on: 26/03/2024.

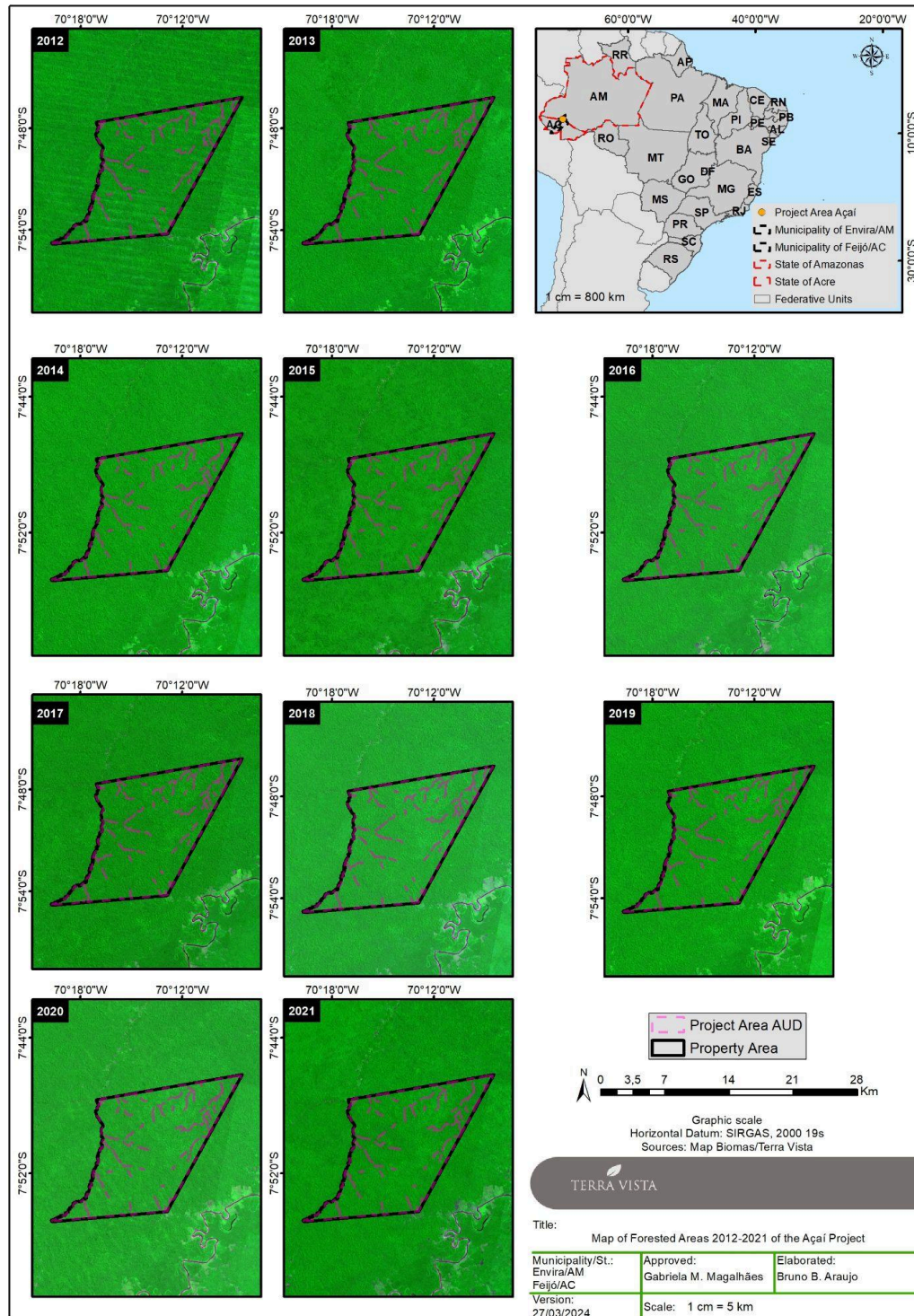


Figure 1.1. Forested areas within the project areas from 2012 to 2022.

The project area meets the definition accepted internationally by FAO/UNFCCC⁹ and the specific thresholds for the Brazilian Amazon: a minimum area of 0.5 hectares, a minimum height of 5 m, and a minimum canopy cover of 10%. As described in section 1.14, the entire project area is composed of Ombrophylous Forest.

1.4.3 Transfer project eligibility

Not applicable.

1.5 Project Design

The Açaí REDD+ project was designed as a "Single location or installation".

1.5.1 Grouped project design

The Açaí REDD+ project is not a grouped project.

1.6 Project Proponent

Table 1.3. Contact information for the proponents of the Açaí REDD+ project.

Organization name	Terra Vista Gestora de Recursos Ltda.
Contact person	Guilherme Rosseto Nunes de Oliveira
Title	Legal Director
Address	Rua Gumerindo Saraiva, nº 54, Sala 04, Jardim Europa, São Paulo, 01449-070 (postal code), Brazil
Telephone	+55 11 4883-1165
Email	projetos@terravista.eco.br

Organization name	J O Cruz & O B Oliveira Participações Ltda.
Contact person	Jose De Oliveira Cruz
Title	landowner

⁹ FAO - Forests and Climate Change Working Paper 17. Available at: <<https://www.fao.org/3/CA0176EN/ca0176en.pdf>>. Accessed on: 26/03/2024.

Address	Rua Sergipe no 358, Sala 02 Térreo/Fundos, Centro, Zip Code 69.900-072, Rio Branco/AC.
Telephone	+55 68 8401-5781
Email	edson.cruz@grupojcruz.com.br

1.7 Other Entities Involved in the Project

Not applicable. No other entities are involved in the Açaí REDD+ project.

1.8 Ownership

The project encompasses a property registered under enrollment number 670 before the Real Estate Registry Office of Envira/AM. This property is owned by J O Cruz & O B Oliveira Participações Ltda. They entered into a public deed of sale and purchase with the heirs of Adrião Batalha on October 9th, 2023, in relation to the Purchase and Sale Agreement signed on June 6th, 2023. This deed was recorded by the 1st Notary Office of Autazes/AM, in book 27, pages 181/184, as pertaining to an area of 16,229.0943 hectares in the mentioned enrollment, which is currently undergoing registration.

The proponents signed a Partnership Agreement for the development of this project prior to the acquisition of the area, while the area was being regularized by Adrião Batalha's heirs. In the mentioned document, it is stated that the responsibility for the execution and results of the project is exclusively held by Terra Vista Gestora de Recursos Ltda., which is responsible for the development, implementation, and monitoring of all project stages. Furthermore, the document specifies that Terra Vista Gestora de Recursos Ltda. is the project administrator, and responsible for its administration in relation to Verra. Terra Vista holds no ownership to the land but only to the carbon credits.

According to the VCS Program requirements, the project complies with item 4 of section 3.7.1 of VCS Standard: *"4) Project ownership arising by virtue of a statutory, property or contractual right in the land, vegetation or conservational or management process that generates GHG emission reductions and/or removals (where the project proponent has not been divested of such project ownership)"*.

1.9 Project Start Date

Table 1.4. Project start date.

Project start date	22-February-2022
Justification	The start date of the project is February 22, 2022, which is the date of the contract signing between Terra Vista Gestora de Recursos Ltda. and J O Cruz & O B Oliveira Participações Ltda. According to the VCS Program requirements, the project start date of an AFOLU project is the date on which activities that lead to the generation of reductions or removals are implemented. In the Açaí REDD+ project, the activities refer to protection and conservation of the area.

1.10 Project Crediting Period

Table 1.5. Project crediting period.

Crediting period	The crediting period of the Açaí REDD+ project will be from 22-February-2022 (start date) to 22-February-2062, contemplating a 40-year and one day period.
Start and end date of first or fixed crediting period	22-February-2022 to 22-February-2062.

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ $< 300,000$ tCO₂e/year (project)Inse

☐ $\geq 300,000$ tCO₂e/year (large project)

Table 1.6. Annual GHG emission removals of the project area in the crediting period.

Calendar year of crediting period	Total Estimated net GHG emissions reductions (tCO ₂ e)
22-Feb-2022 to 31-Dec-2022	202,078.23
01-Jan-2023 to 31-Dec-2023	240,689.28
01-Jan-2024 to 31-Dec-2024	246,565.37
01-Jan-2025 to 31-Dec-2025	252,441.45
01-Jan-2026 to 31-Dec-2026	258,317.54
01-Jan-2027 to 31-Dec-2027	264,193.62
01-Jan-2028 to 31-Dec-2028	270,069.71

01-Jan-2029 to 31-Dec-2029	275,945.79
01-Jan-2030 to 31-Dec-2030	281,821.88
01-Jan-2031 to 31-Dec-2031	287,697.97
01-Jan-2032 to 31-Dec-2032	289,436.94
01-Jan-2033 to 31-Dec-2033	290,488.59
01-Jan-2034 to 31-Dec-2034	291,540.25
01-Jan-2035 to 31-Dec-2035	292,591.91
01-Jan-2036 to 31-Dec-2036	293,643.56
01-Jan-2037 to 31-Dec-2037	294,695.22
01-Jan-2038 to 31-Dec-2038	295,746.88
01-Jan-2039 to 31-Dec-2039	296,798.53
01-Jan-2040 to 31-Dec-2040	297,850.19
01-Jan-2041 to 31-Dec-2041	298,901.85
01-Jan-2042 to 31-Dec-2042	299,051.67
01-Jan-2043 to 31-Dec-2043	299,051.67
01-Jan-2044 to 31-Dec-2044	299,051.67
01-Jan-2045 to 31-Dec-2045	299,051.67
01-Jan-2046 to 31-Dec-2046	299,051.67
01-Jan-2047 to 31-Dec-2047	299,051.67
01-Jan-2048 to 31-Dec-2048	299,051.67
01-Jan-2049 to 31-Dec-2049	299,051.67
01-Jan-2050 to 31-Dec-2050	299,051.67
01-Jan-2051 to 31-Dec-2051	299,051.67
01-Jan-2052 to 31-Dec-2052	299,051.67
01-Jan-2053 to 31-Dec-2053	299,051.67
01-Jan-2054 to 31-Dec-2054	299,051.67
01-Jan-2055 to 31-Dec-2055	167,090.92
01-Jan-2056 to 31-Dec-2056	78,586.14
01-Jan-2057 to 31-Dec-2057	63,864.75
01-Jan-2058 to 31-Dec-2058	58,217.77
01-Jan-2059 to 31-Dec-2059	52,341.69
01-Jan-2060 to 31-Dec-2060	46,465.60
01-Jan-2061 to 31-Dec-2061	40,589.52

01-Jan-2062 to 22-Feb-2062	5,663.35
Total estimated ERRs during the fixed crediting period	9,922,006.23
Total number of crediting years	40
Average annual ERs	248,050.16

1.12 Description of the Project Activity

The Açaí REDD+ project is not located within a jurisdiction covered by a jurisdictional REDD+ program. The main objective of the project is to promote actions aimed at reducing greenhouse gases (GHG) emissions from unplanned deforestation. In this way, some activities will be proposed, such as the improvement of heritage surveillance, remote monitoring of forest cover, monitoring and deepening of biodiversity studies, and activities to intensify the engagement of communities with the preservation of the forest and forest resources.

Thus, to ensure positive benefits for the three main scopes: climate, community, and biodiversity, the project activities were defined based on the reality and local context, always aiming at continuous improvement. The successful implementation of the project's activities guarantees the commercialization of carbon credits, which return as financial resources for the continuation of social development activities and natural resources conservation.

1.13 Project Location

The Açaí REDD+ project is situated on a property spanning 16,228.39 hectares, stretching across two states in Brazil: the municipality of Envira in the southwest region of Amazonas State, and the municipality of Feijó located in the central region of Acre State (Figure 1.2). The project area is inserted in the geographical coordinates 70°19'39.48"W to 70°8'32.04"W and 7°46'13.57" S to 7°54'47,59"S and encompasses 15,715.82 ha destined for Avoided UnPlanned Deforestation (AUD).

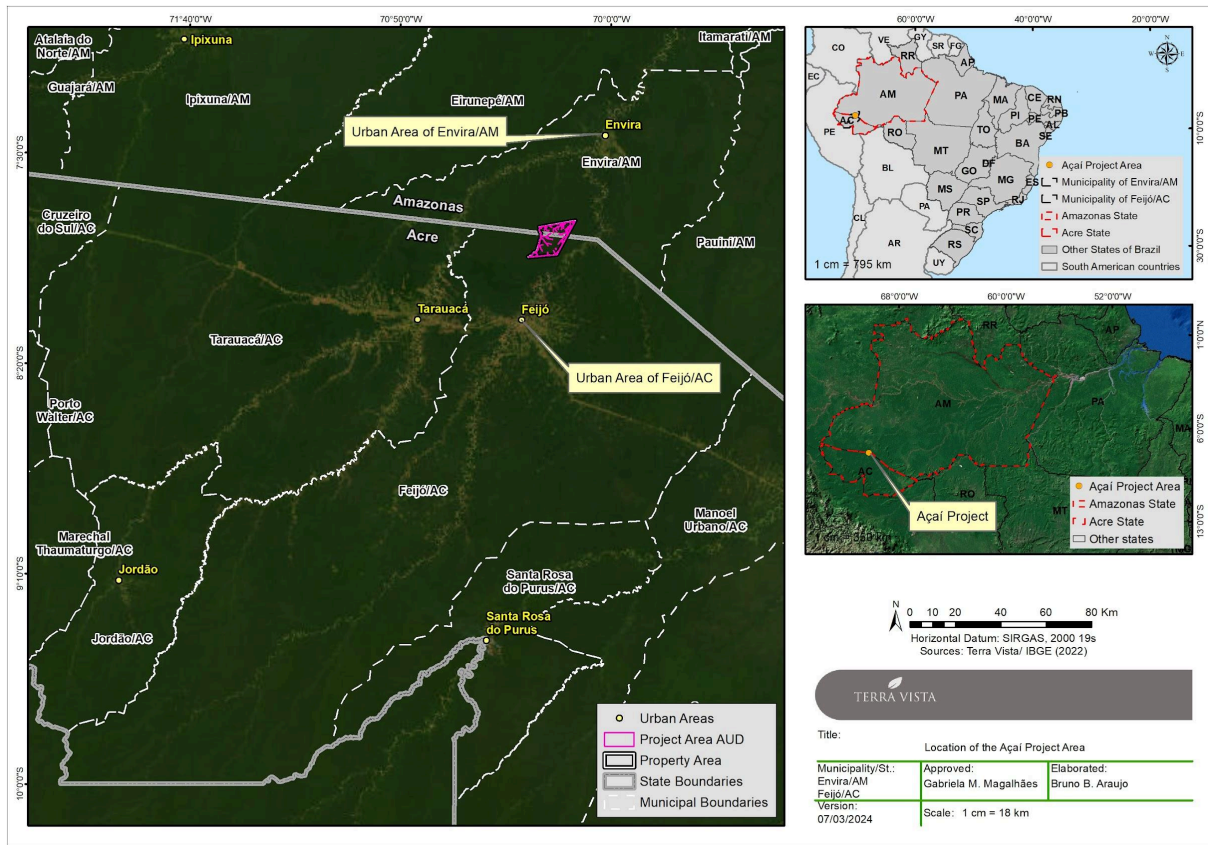


Figure 1.2. Location of the Açaí project in the states of Amazonas and Acre, Brazil.

1.14 Conditions Prior to Project Initiation

The Açaí REDD+ project will be implemented with the aim of reducing greenhouse gas emissions related to deforestation and forest degradation, as required by scope 14 of the VCS. The project's activities include protecting forest areas threatened by deforestation and degradation, which have been classified as forest for at least 10 years before the project's start date. The project area is not intended to generate greenhouse gas emissions for the purpose of subsequent reduction, removal, or destruction.

As presented in section 1.13, the project is located in the municipalities of Envira - AM and Feijó - AC. The municipality of Envira-AM has an area of 7,505.79 km² and it is mostly composed of natural Forest Formations (94.75%) with a small portion of pasture areas (3.61%) and other land cover classes (Figure 1.3). Envira borders the municipality of Feijó-AC, which presents an area of 27.976,87 km². About 93.4% of the Feijó's territory is made up of forest formations and 6.08% of areas dedicated to pastures (Figure 1.4). The property where the Açaí REDD+ project is also mostly

composed of Forest Formations (99.93%) with a small portion of pasture areas (0.06%) and Floodable Forests (0.01%) (Figure 1.4).

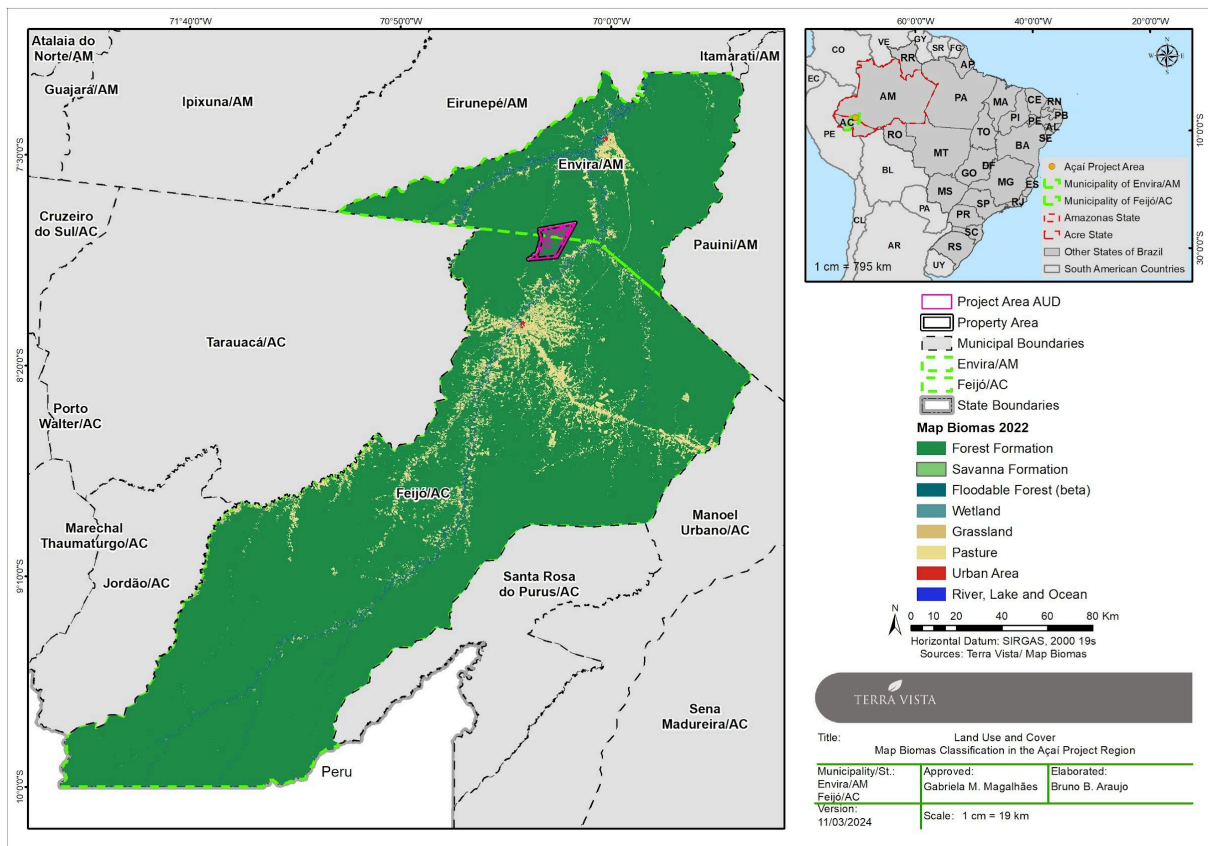


Figure 1.3. Land Use and Land Cover in the municipalities of Envira - AM and Feijó - AC, Brazil where the Açai REDD+ project is located.

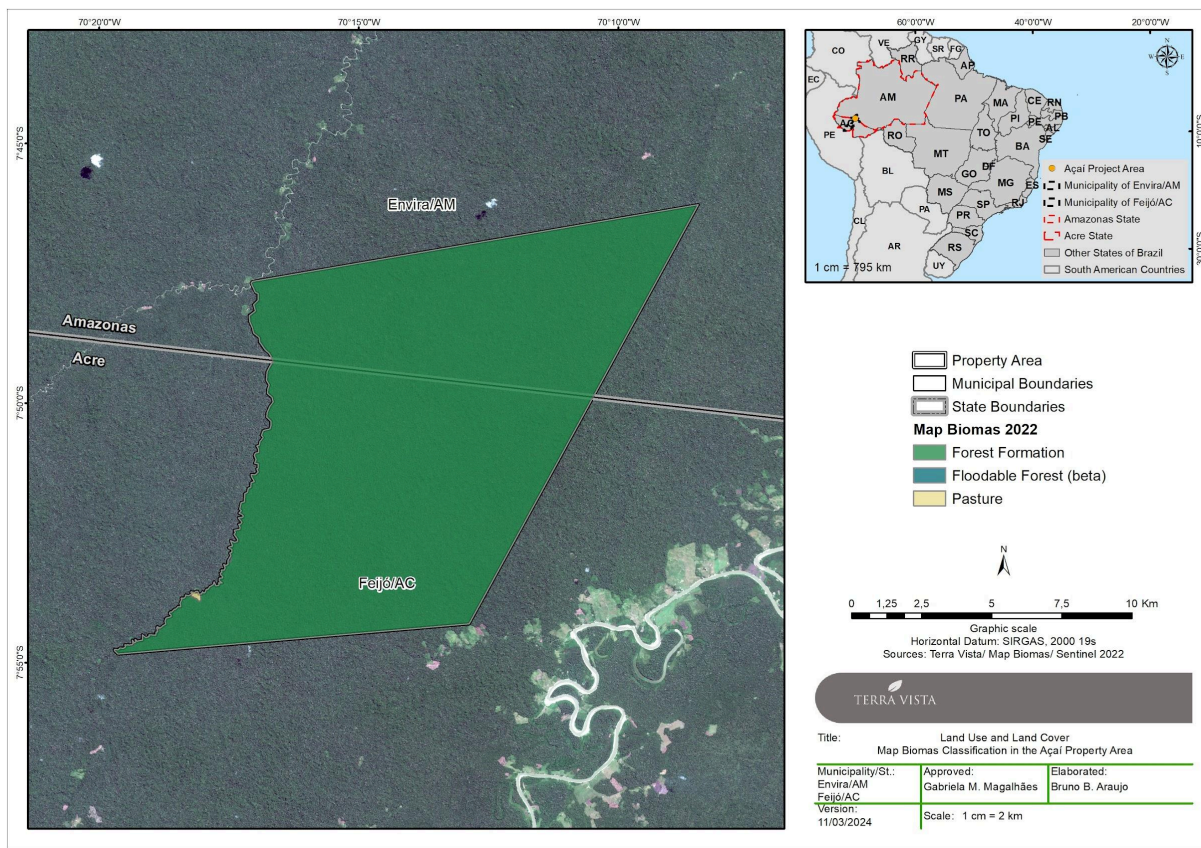


Figure 1.4. Land Use and Land Cover in the Açaí REDD+ project area, Brazil.

Geology, Geomorphology and Topography

The state of Amazonas is characterized by an extensive Phanerozoic sedimentary cover (covering the last 542 million years) deposited on a Precambrian rocky substrate (spanning from 4.5 billion years ago to the Phanerozoic). The Açaí REDD+ project area is inserted mainly in the Amazonas-Solimões Province on the Solimões Formation (Figure 1.5). The Solimões Formation emerged in the Cenozoic Era around 65.5 million years ago, resulting from a Plio-Pleistocene deposition and is composed of a set of rocks and minerals, such as claystones with intercalations of siltstones, sandstones, limestones and material carbonaceous, of fluvial and fluviolacustrine origin¹⁰. Additionally, the region features Holocene Terraces, which are fluvial plain deposits composed of lenticular channel bottom gravels, unconsolidated quartz sands, silts and overflow clays¹¹.

¹⁰ IBGE. Geologia do Estado do Amazonas. Available at: https://geoftp.ibge.gov.br/informacoes_ambientais/geologia/levantamento_geologico/mapas/unidades_da_federacao/am_geologia.pdf. Accessed on: 04/03/2024.

¹¹ IBGE. Geologia do Estado do Amazonas. Available at: https://geoftp.ibge.gov.br/informacoes_ambientais/geologia/levantamento_geologico/mapas/unidades_da_federacao/am_geologia.pdf. Accessed on: 04/03/2024.

Furthermore, there are Holocene Alluvial Deposits, which consist of conglomerates, sandstones and claystones¹².

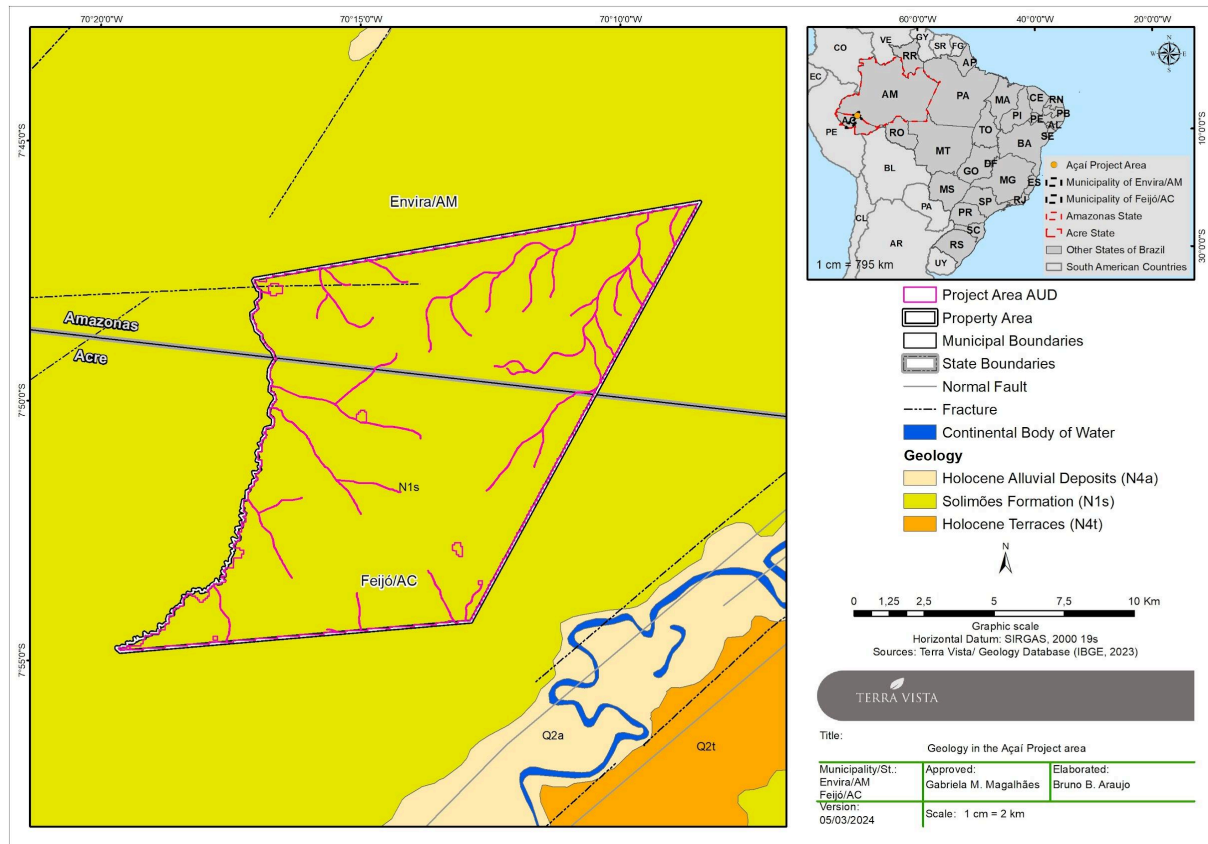


Figure 1.5. Geology of the region where the Açaí REDD+ project is located.

The Amazon has reliefs of different shapes that are defined from the variation of altitudes of its plains. The relief of the Açaí REDD+ project area is quite uniform, without accentuated topographical contrasts, formed on Phanerozoic Sedimentary Covers and Basins (Figure 1.6a). The Açaí REDD+ project area lies mostly on the Juruá-Iaco Depression (Figure 1.6b). In the region there is also the Amazon Plain, the largest plain in the world and is represented by the floodplain area of the Amazon River and its main tributaries. In addition, the predominant relief of the project area is Dissected with Convex Top (Figure 1.6c).

¹² IBGE. Geologia do Estado do Amazonas. Available at: https://geoftp.ibge.gov.br/informacoes_ambientais/geologia/levantamento_geologico/mapas/unidades_da_federacao/am_geologia.pdf. Accessed on: 04/03/2024.

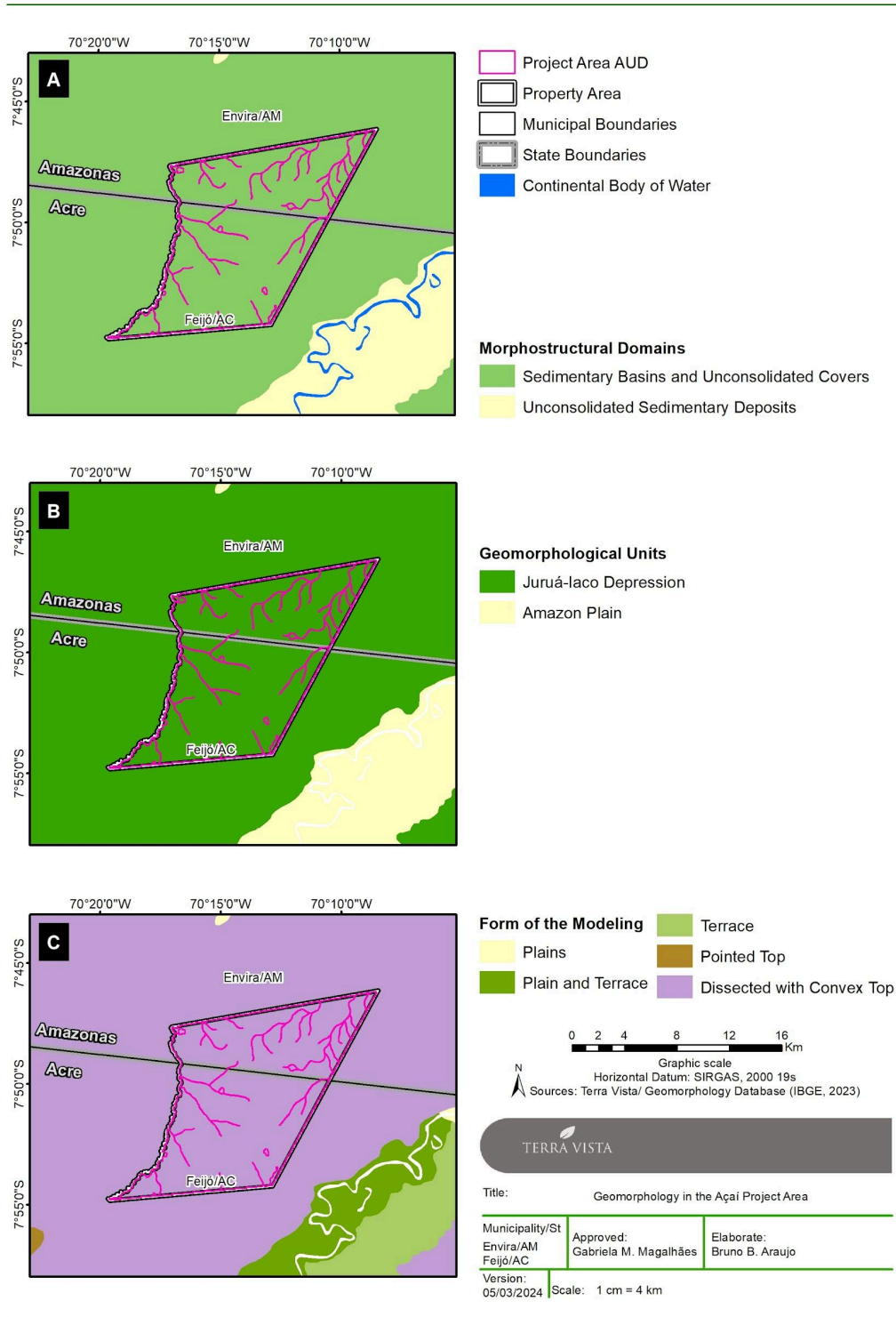


Figure 1.6. Geomorphology of the region where the Açaí REDD+ project is located¹³.

¹³ IBGE. Geomorphology. Available at:

<<https://www.ibge.gov.br/geociencias/informacoes-ambientais/geomorfologia/10870-geomorfologia.html>>. Accessed on: 04/03/2024.

Soil

Natural factors such as climate, relief, time, and parent material, combined in different intensities, form the various soil types observed in the region of the Açaí REDD+ project. The predominant soil type in the project area is Chromic Luvisol - TCo (79.75%), with a minor presence of Haplic Luvisol - TXo (20.25%) (Figure 1.7).

Luvisols (Alfisols) are mineral soils, non-hydromorphic, with a textural B horizon with high activity clay and high base saturation (eutrophic), immediately below the A or E horizon; they are usually shallow and vary from well to imperfectly drained¹⁴. The textural B horizon has a higher clay content than the more superficial horizons (A or E), which hinders water infiltration, allowing prolonged waterlogging after the rainy season. The Chromic character refers to the bright colors in the B horizon and, although there is no direct agronomic importance, it indicates that these soils do not have restriction of use due to imperfect drainage¹⁵. Conversely, the term "haplic" denotes a lack of further or substantial characterization¹⁶, so a haplic luvisol lacks significant diagnostic features or characteristics associated with other subtypes of luvisols.

¹⁴ EMBRAPA. 2018. Brazilian Soil Classification System. Humberto Gonçalves dos Santos ... [et al.]. – 5. ed., rev. e ampl. – Brasília, DF : Embrapa. Available at: <https://www.embrapa.br/busca-de-publicacoes/-/publicacao/1094001/brazilian-soil-classification-system>. Accessed on: 03/04/2024.

¹⁵ EMBRAPA. 2018. Sistema Brasileiro de Classificação de Solos / Humberto Gonçalves dos Santos ... [et al.]. – 5. ed., rev. e ampl. – Brasília, DF : Embrapa. Available at: <https://www.embrapa.br/solos/sibcs>. Accessed on: 03/04/2024.

¹⁶ FAO - Annex 3 - Qualifiers (formative elements for naming soil units). Available at: <https://www.fao.org/3/Y1899E/y1899e17.htm>. Accessed on: 03/04/2024.

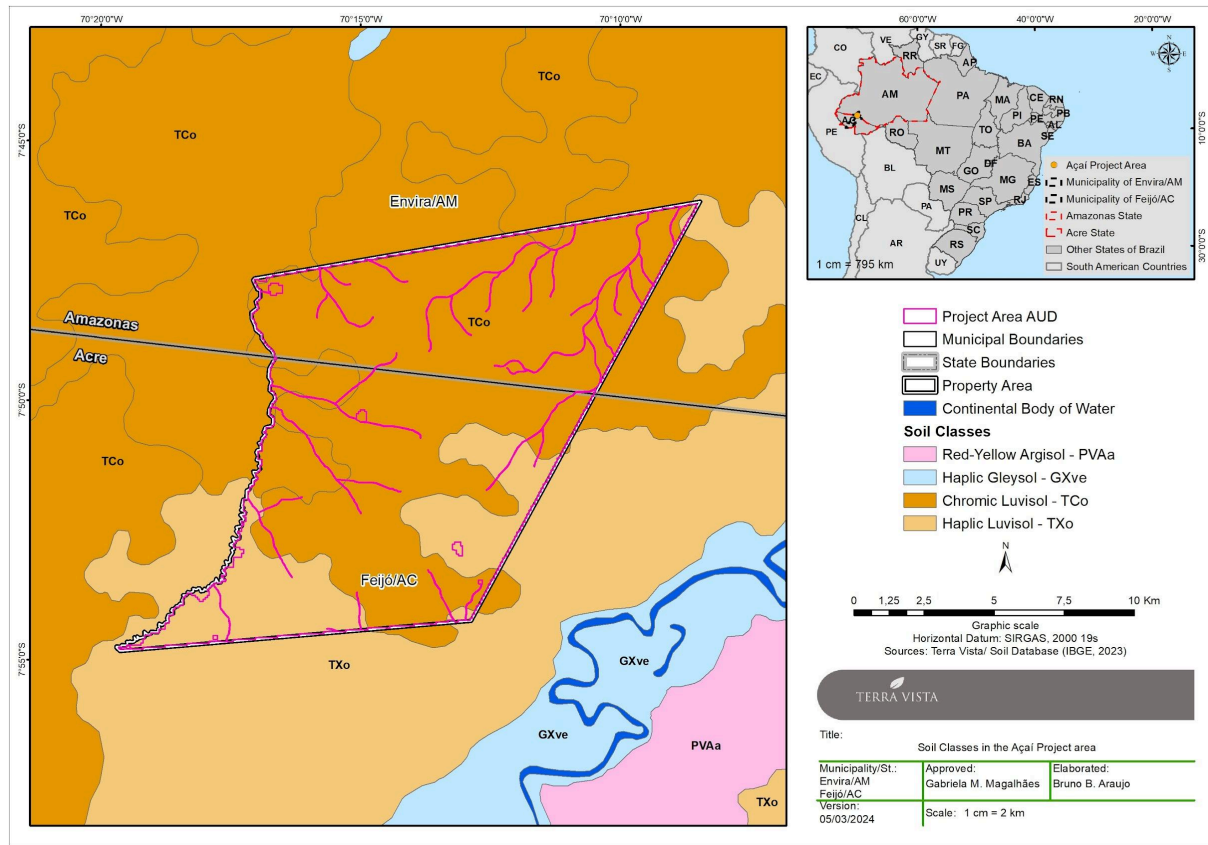


Figure 1.7. Soil classes in the region where the Açaí REDD+ project is located.

Climate

The climate in the region is classified as humid tropical with a short dry season, type “Am” in the Köppen classification (1948)¹⁷. The rainy season typically extends from December to March, peaking in rainfall in January, while the dry season spans from June to August (Figure 1.8)¹⁸. The hottest period of the year is concentrated in the months of November and February, while the lowest temperatures were registered in the months of June to September (Figure 1.9)¹⁹.

¹⁷ Köppen, W. 1948. Köppen climate classification. Available at: <<https://www.britannica.com/science/Koppen-climate-classification>>. Accessed on: 14/03/2023.

¹⁸ Climate Hazards Group (CHIRPS). Available at: <<https://chc.ucsb.edu/data/chirps>>. Accessed on: 07/02/2024.

¹⁹ MODIS. Available at: <<https://doi.org/10.5067/MODIS/MOD11A2.061>>. Accessed on: 07/02/2024.



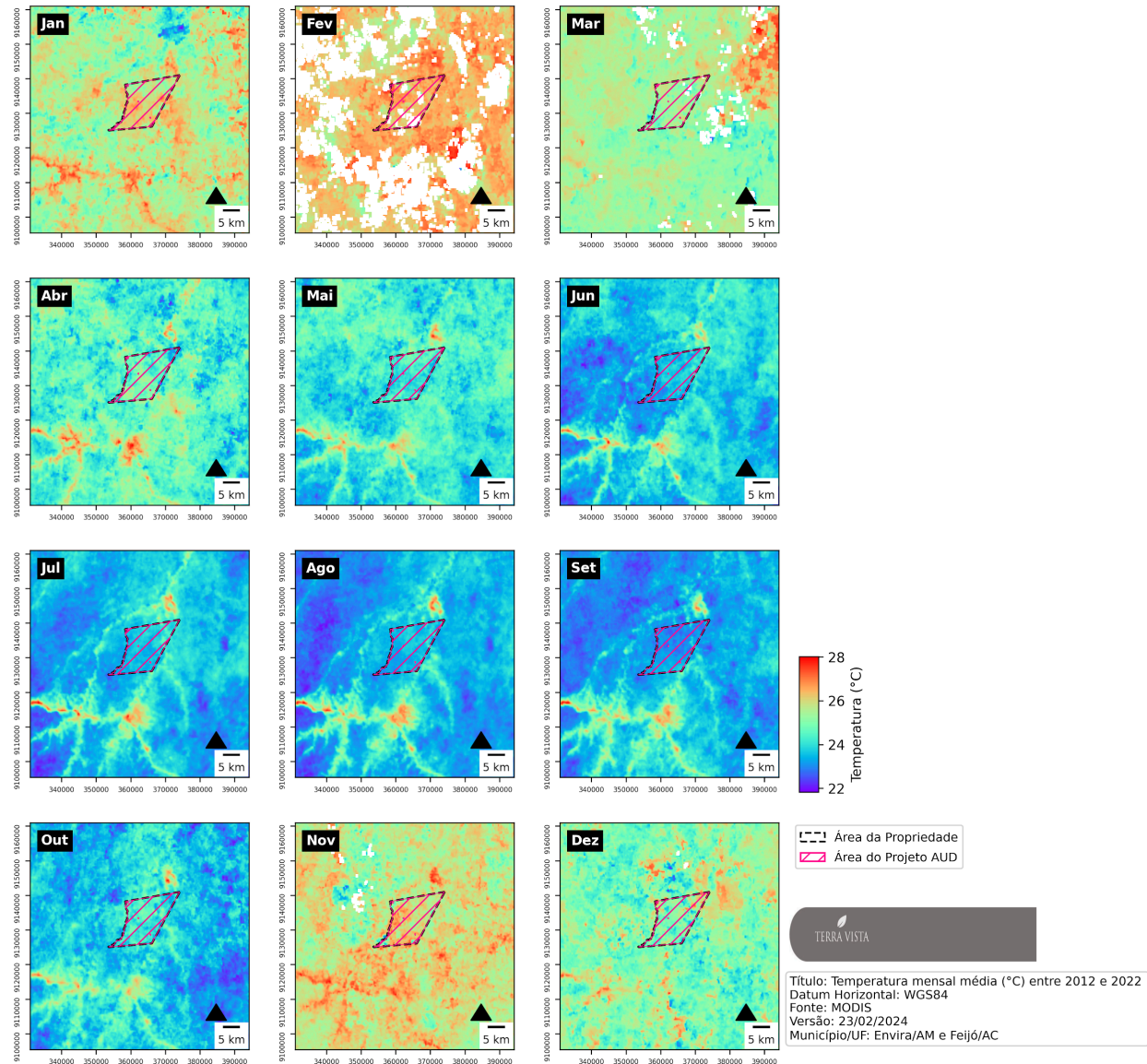


Figure 1.9. Average monthly temperature between 2012-2022 in the region of the Açaí REDD+ project.

Hydrology

The Açaí REDD+ project is located in the largest hydrographic basin in the world, the Amazon Basin, more specifically in the basin of the Juruá, Envira and Tarauacá rivers^{20 21} (Figure 1.10). The

²⁰ FBDS - Brazilian Foundation for Sustainable Development. Available at: <<https://geo.fbds.org.br/AM/ENVIRA/HIDROGRAFIA/>>. Accessed on: 07/02/2024.

²¹ FBDS - Brazilian Foundation for Sustainable Development. Available at: <<https://geo.fbds.org.br/AC/FEIJO/HIDROGRAFIA/>>. Accessed on: 07/02/2024.

Amazon Basin has an estimated area of 5.8 million km², of which 3.9 million km² are in Brazilian territory²², and is formed by several secondary basins that are named according to the main tributary that compose it.

The Juruá River occupy and area of 250.000 km² ²³ and is considered an important tributary of the Amazon hydrographic basin, rising in Peru and running through the states of Acre and Amazonas for a length of 3,280 km until it flows into the right bank of the Solimões River²⁴. In general, the Juruá River floods its plains from December to May, and in its lower course the flooding can last until August, due to the damming effect of the Solimões/Amazonas²⁵.

Envira is a binational river that flows through the states of Acre and Amazonas. It originates in the Contamana mountain range, in Peruvian territory²⁶, and covers a total area of 54,616.49 km² ²⁷. The Tarauacá River is the most important tributary of the Juruá River and one of the main watercourses in the central region of Acre State²⁸. With a sinuous to meandering flow pattern, interspersed with short straight sections, it covers a total area of 23,306.27 km² ²⁹. The river maintains a minimum depth of 1.20 meters and navigability from its mouth to the confluence with the Jordão River near the Peruvian border³⁰. Both the Tarauacá and the Envira rivers are plain rivers, thus they embody typical traits of gently sloping streams, characterized by consistently turbid waters laden with sediments leached from the clayey soils along their banks. Hence, they feature muddy waters, spanning approximately 100 meters in width, and undergo shifts in their bed positions with each flood season³¹.

²² ICMBIO. 2010. Plano de Manejo Participativo da Reserva Extrativista Arapixi. Volume I – Diagnóstico e Caracterização

²³ Bacia do Rio Juruá - Levantamento Etnoecológico das Terras Indígenas do Complexo. Available at: <http://cggamgati.funai.gov.br/files/2514/8829/1721/jurua_pt.pdf>. Accessed on: 04/03/2024.

²⁴ Costa, A. C. S.; Souza, L. P. Delgado, R. C. Gomes, F. A. 2012. Períodos de cheia e vazante do rio Juruá na região de Cruzeiro do Sul, Acre, ENCICLOPÉDIA BIOSFERA, Centro Científico Conhecer - Goiânia, v.8, N.14; p. 1 3 4 3. Available at: <<https://conhecer.org.br/ojs/index.php/biosfera/article/view/3957>>. Accessed on: 04/03/2024.

²⁵ WCS - Wildlife Conservation Society. 2019. Available at: <<http://pt.aguasamazonicas.org/basins/main-subbasins/jurua/>>. Accessed on: 04/03/2024.

²⁶ Rio Envira. Available at: <https://pt.wikipedia.org/wiki/Rio_Envira>. Accessed on: 04/03/2024.

²⁷ Bacias e Divisões Hidrográficas do Brasil | 2021. Available at: <<https://www.ibge.gov.br/geociencias/cartas-e-mapas/informacoes-ambientais/31653-bacias-e-divisoes-hidrograficas-do-brasil.html>>. Accessed on: 04/03/2024.

²⁸ Bacia do Rio Juruá - Levantamento Etnoecológico das Terras Indígenas do Complexo. Available at: <http://cggamgati.funai.gov.br/files/2514/8829/1721/jurua_pt.pdf>. Accessed on: 04/03/2024.

²⁹ Bacias e Divisões Hidrográficas do Brasil | 2021. Available at: <<https://www.ibge.gov.br/geociencias/cartas-e-mapas/informacoes-ambientais/31653-bacias-e-divisoes-hidrograficas-do-brasil.html>>. Accessed on: 04/03/2024.

³⁰ Bacia do Rio Juruá - Levantamento Etnoecológico das Terras Indígenas do Complexo. Available at: <http://cggamgati.funai.gov.br/files/2514/8829/1721/jurua_pt.pdf>. Accessed on: 04/03/2024.

³¹ Bacia do Rio Juruá - Levantamento Etnoecológico das Terras Indígenas do Complexo. Available at: <http://cggamgati.funai.gov.br/files/2514/8829/1721/jurua_pt.pdf>. Accessed on: 04/03/2024.

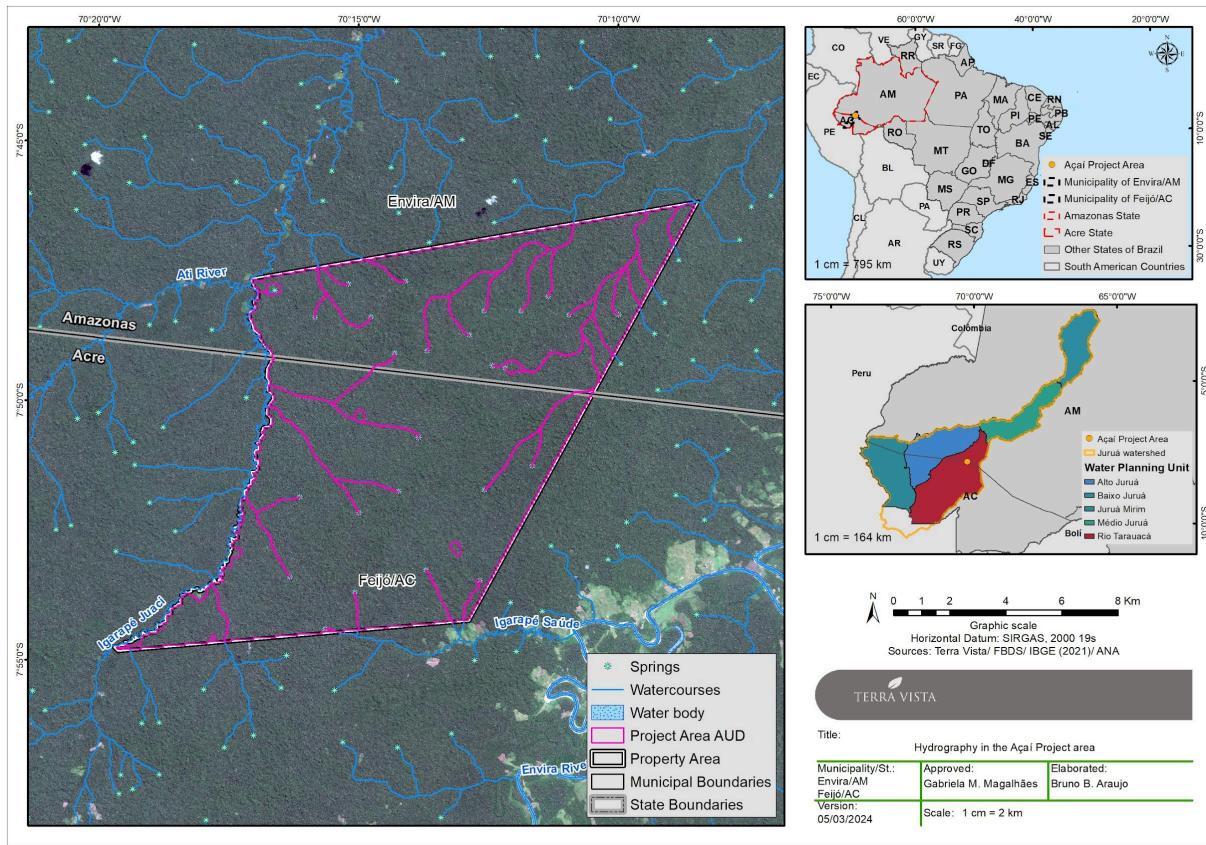


Figure 1.10. Hydrography in the region where the Açaí REDD+ project is located.

Vegetation Types

The Açaí REDD+ project is situated within is located in the Amazon Biome and it is predominantly characterized by the presence of the following phytophysionomies: Lowland Open Ombrophyllous Forest (Lo), Alluvial Open Ombrophyllous Forest (Ao) and Alluvial Dense Ombrophyllous Forest (Ad), according to IBGE^{32 33} (Figure 1.11 and Table 1.7). The vegetation in the Açaí project area is the Lowland Open Ombrophyllous Forest, a formation that occurs at altitudes ranging from 5 to 100m, and is composed of more widely spaced trees, with a low-density shrub stratum and characterized by phanerophytes and woody lianas³⁴.

³² Vegetation of the state of Amazonas. Available at: http://geoftp.ibge.gov.br/informacoes_ambientais/vegetacao/mapas/unidades_da_federacao/am_vegetacao.pdf. Accessed on: 20/01/2024.

³³ IBGE. 2021. Mapa de vegetação Vegetação 1:250.000. Available at: https://geoftp.ibge.gov.br/informacoes_ambientais/vegetacao/vetores/escala_250_mil/versao_2021/. Accessed on: 20/01/2024.

³⁴ IBGE 2012. Manual Técnico da Vegetação Brasileira. 2aed. Available at: <https://biblioteca.ibge.gov.br/index.php/biblioteca-catalogo?view=detalhes&id=263011> Accessed on: 13/10/2022.

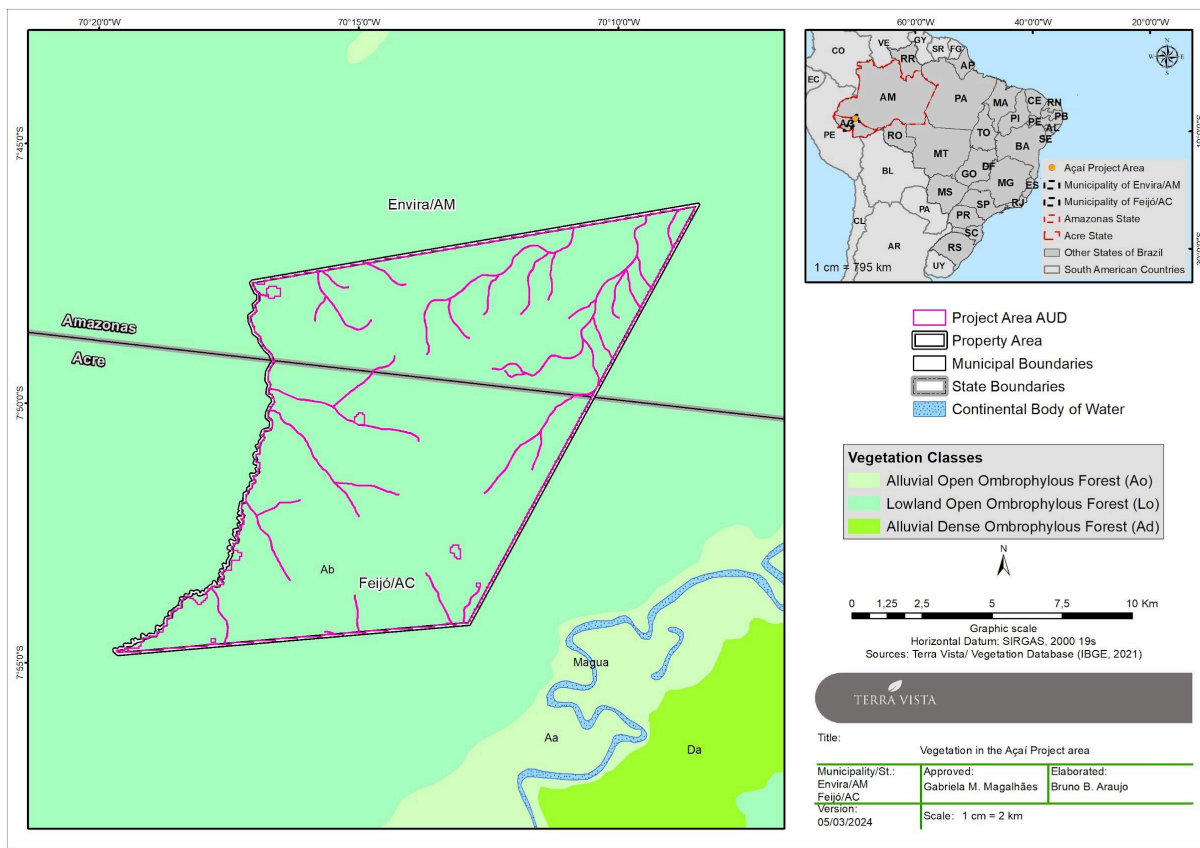


Figure 1.11. Type of vegetation in the Açaí REDD+ project region.

Table 1.7. Types of vegetation found in the Açaí REDD+ project area³⁵.

Vegetation Type	Area (ha)	Area (%)
Lowland Open Ombrophylous Forest	15,715.82	100%
Total	15,715.82	100%

Protected and Priority Areas for Conservation

Conservation Units

In the region close to the Açaí REDD+ project, there are many Conservation Units (i.e. Protected Areas) for Sustainable Use, which aim to make nature conservation compatible with the sustainable use of its natural resources (Figure 1.12). Additionally, in the region there is also a

³⁵ Vegetation of the state of Amazonas. Available at: http://geofp.ibge.gov.br/informacoes_ambientais/vegetacao/mapas/unidades_da_federacao/am_vegetacao.pdf. Accessed on: 20/01/2024.

Conservation Units for Full Protection, whose purpose is to preserve nature, admitting only the indirect use of its natural resources (except for the cases foreseen in Law No. 9,985/2000³⁶), such as the Chandless State Park ('Parque Estadual Chandless') (Figure 1.12).

These protected areas play a crucial role in preserving substantial portions of the region's ecosystems. In the figure below, it is possible to observe that the Açaí project does not overlap with any Conservation Unit of Sustainable Use or Full Protection.

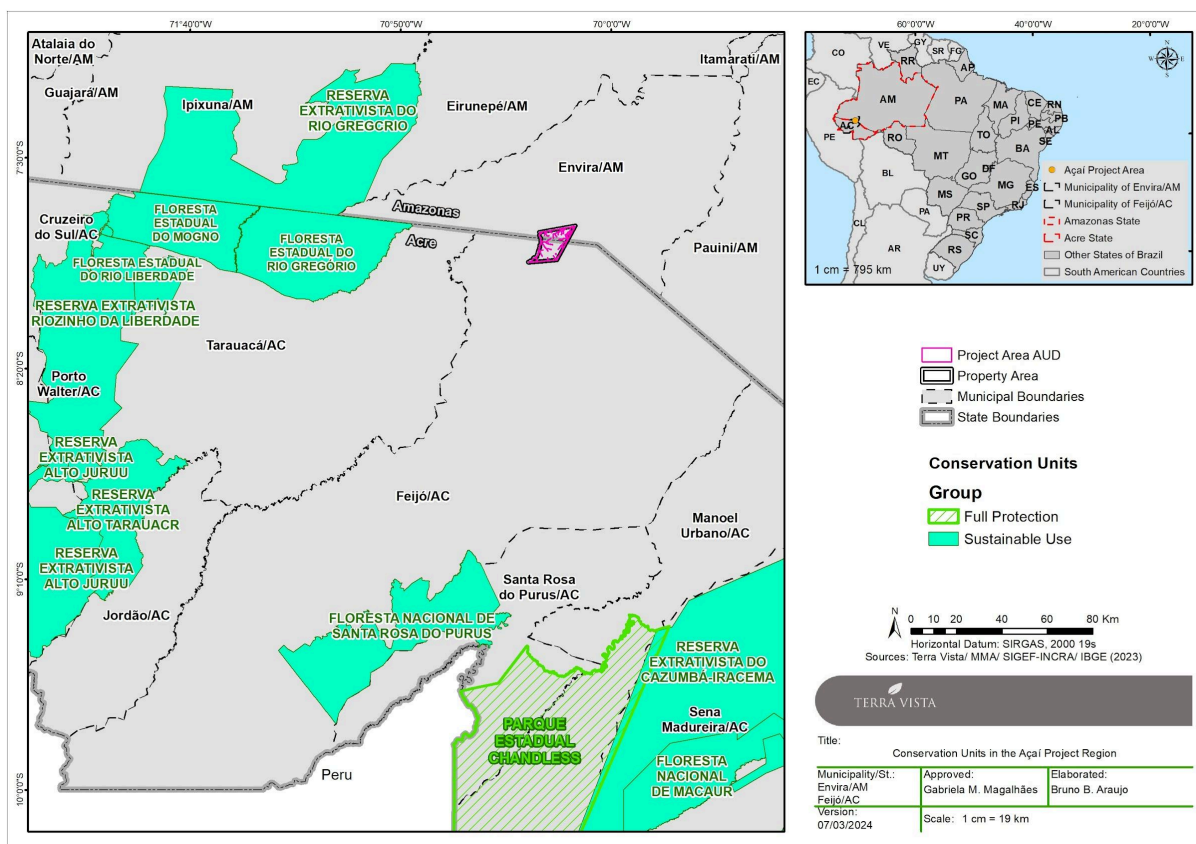


Figure 1.12. Conservation Units (or Protected Areas) in the region where the Açaí REDD+ project is located.

Priority Areas for Conservation

The Priority Areas for Conservation, established by Decree No. 5.092/2004³⁷, are considered a public policy instrument that aims to help in the decision making process, in an objective and

³⁶ Lei no 9.985/2000. Available at: <http://www.planalto.gov.br/ccivil_03/leis/19985.htm>. Accessed on: 07/02/2024.

³⁷ Decreto nº 5.092/2004. Available at: <http://www.planalto.gov.br/ccivil_03/_ato2004-2006/2004/decreto/d5092.htm>. Accessed on: 11/13/2022

participative manner, besides supporting the planning and implementation of actions for the conservation of Brazilian biodiversity³⁸. These areas are not protected by law and, therefore, do not have any restriction on the development of projects within their boundaries.

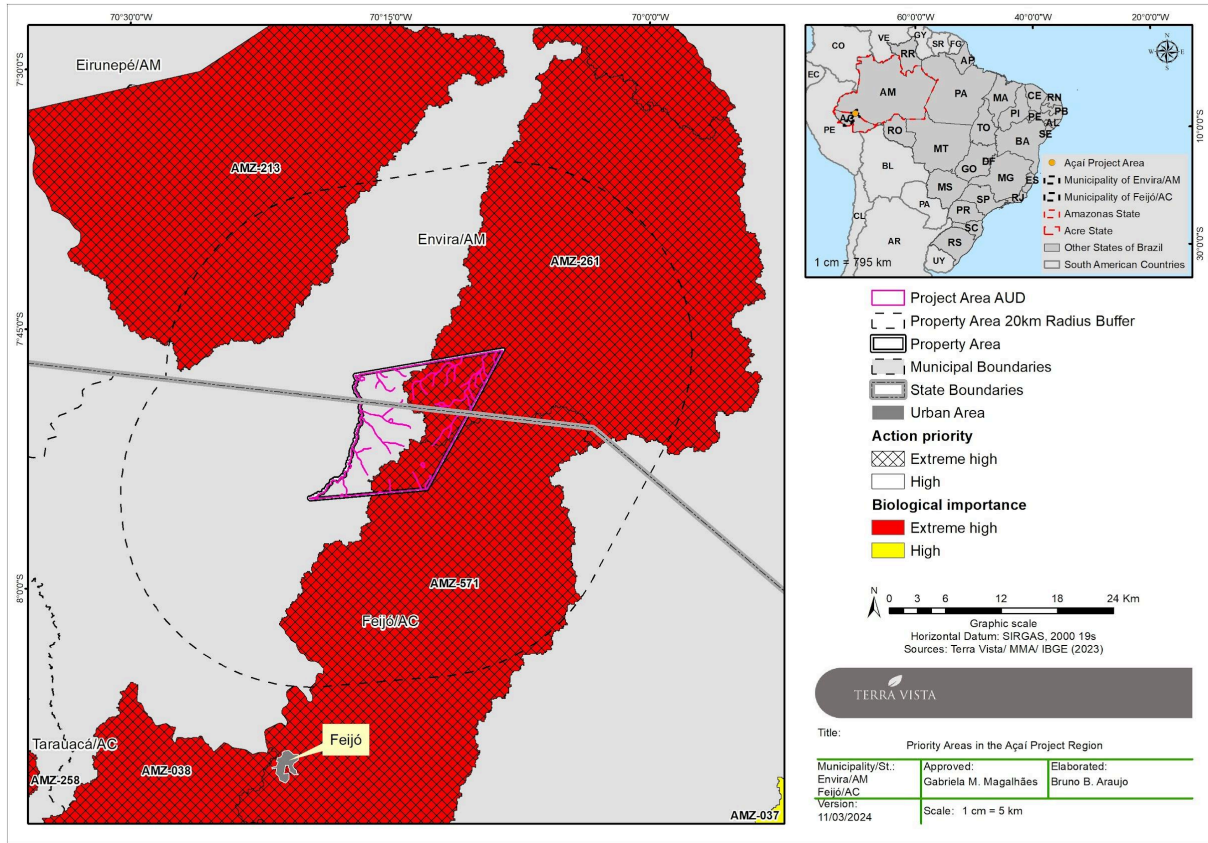
As provided in MMA Ordinance No. 463/2018³⁹, the areas selected for conservation are subject to the formulation and implementation of public policies, programs, projects and activities under the responsibility of the Federal Government aimed at: In situ conservation of biodiversity; Sustainable use of biodiversity components; Sharing of benefits derived from access to genetic resources and associated traditional knowledge; Research and inventories on biodiversity; Recovery of degraded areas and overexploited or endangered species; and Economic valuation of biodiversity.

Also, according to the aforementioned Ordinance, the actions will be implemented considering two classes of biological importance and prioritization of action: Classes of Biological Importance (extremely high, very high, high, and insufficiently known) and Classes of Priority for Action (extremely high, very high, and high).

According to MMA (2018), the Açaí REDD+ project area is close to overlaps areas of “Extremely High” biological importance and conservation priority action (AMZ-261 and AMZ-571) and it is also close to other areas of “Extremely High” relevance (e.g., AMZ-213 and AMZ-038) (Figure 1.13).

³⁸ MMA - Ministério do Meio Ambiente e Mudança do Clima. Áreas prioritárias para Biodiversidade. Available at: <<https://www.gov.br/mma/pt-br/assuntos/ecossistemas-1/areas-prioritarias-para-biodiversidade/areas-prioritarias>>. Accessed on: 14/03/2023.

³⁹ MMA Portaria nº 463/2018. Available at: <https://www.in.gov.br/materia/-/asset_publisher/Kujrw0TZC2Mb/content/id/55881195/do1-2018-12-19-portaria-n-463-de-18-de-dezembro-de-2018-55880954>. Accessed on: 07/02/2024.



Key Biodiversity Area (KBA)

In addition to being close to Protected and Priority Areas for Conservation, the Açaí REDD+ project area is located near a Key Biodiversity Area (KBA) named Tabocais, according to the IBAT (Integrated Biodiversity Assessment Tool)⁴⁰ (Figure 1.14). KBAs comprise an “umbrella” set of internationally recognized priority sites for biodiversity, which meet criteria based on a vulnerability and uniqueness framework defined in terms of threat levels to species and populations. This highlights the importance of the project's region for biodiversity conservation.

⁴⁰ IBAT Proximity Report. Generated under license 31998-59379 from the Integrated Biodiversity Assessment Tool on 04 March 2024 (GMT). Available at: <<http://www.ibat-alliance.org>>. Accessed on: 04/03/2024.

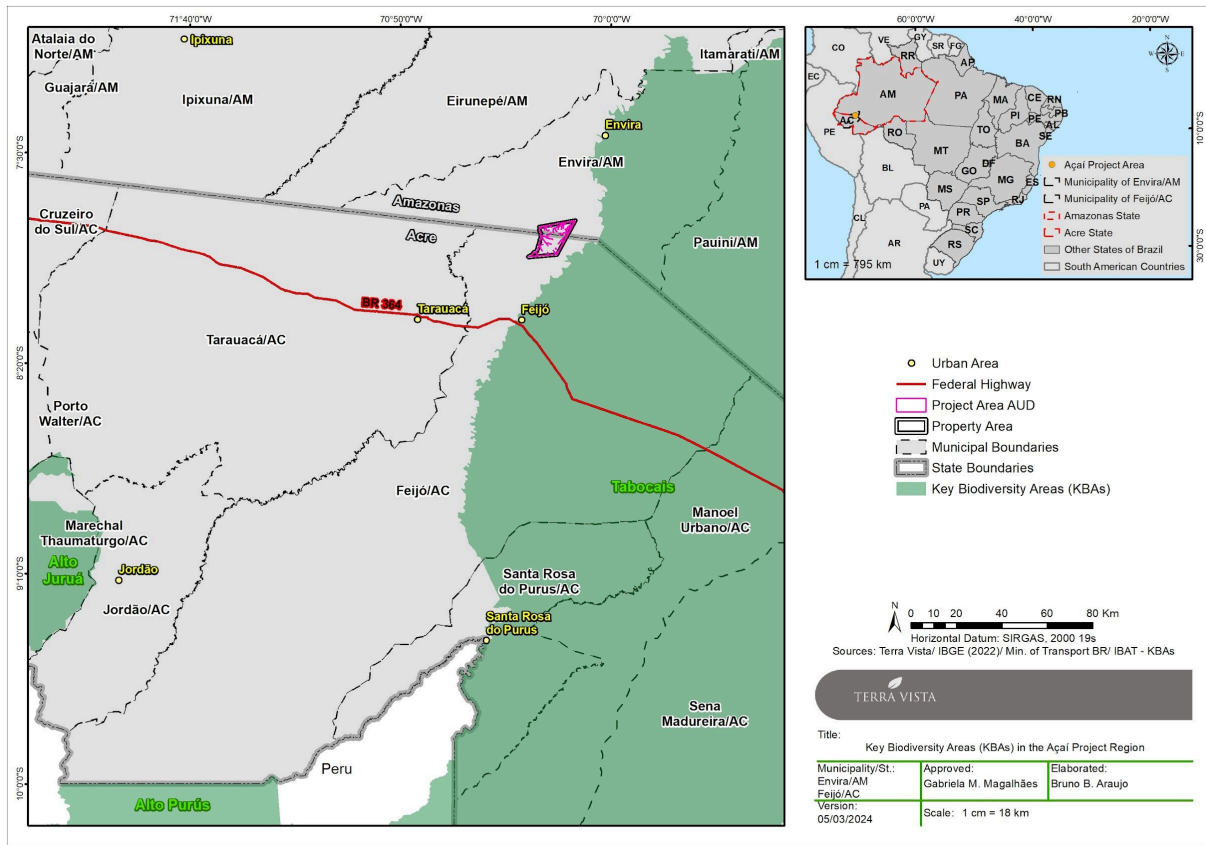


Figure 1.14. Key Biodiversity Areas (KBAs) in the region of the Açaí REDD+ project.

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

Proponents of the Açaí REDD+ project are committed to complying with applicable and relevant national, state and local laws, including statutes and regulatory frameworks.

International agreements:

- Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), 1975;
- Cartagena Protocol on Biosafety to the Convention on Biological Diversity, 2000;
- International Tropical Timber Agreement (ITTA), 1994;
- Convention on Biological Diversity (CBD), 1992;
- United Nations Framework Convention on Climate Change (UNFCCC), 1992;
- United Nations Declaration on the Rights of Indigenous Peoples, 2007;

- Kyoto Protocol, 1997;
- Paris Agreement, 2015.

National laws:

- **Brazilian Federal Constitution of 1988**, under Title VIII, of the social order, in Chapter VI, on the environment, article 225 *"Everyone has the right to an ecologically balanced environment, a good for common use by the people and essential to a healthy quality of life, imposing on the Public Power and the community the duty to defend and preserve it for present and future generations"* and in its 1st paragraph, item III, provides *"to define, in all units of the Federation, territorial spaces and their components to be specially protected, alteration and deletion being permitted only by law, any use that compromises the integrity of the attributes that justify their protection is prohibited"*.

Still in the aforementioned article, the Law provides in its § 4th *"The Brazilian Amazon Forest, the Atlantic Forest, the Serra do Mar, the Mato-Grossense Pantanal and the Coastal Zone are national heritage, and their use will be made, in law, under conditions that ensure the preservation of the environment, including the use of natural resources."*

Art. 68. Permanent ownership is recognized for the remnants of the quilombo communities that are occupying their lands, and the State must issue them the respective titles.

Art. 215. The State will guarantee to all the full exercise of cultural rights and access to sources of national culture, and will support and encourage the appreciation and dissemination of cultural manifestations.

§ 1º The State will protect the manifestations of popular, indigenous and Afro-Brazilian cultures, and those of other groups participating in the national civilizing process.

In chapter II, which deals with social rights, there are labor laws in articles 7, 8, 9, 10 and 11.

- **National Environmental Policy, provided for in Law 6,938 of 1981.** The aforementioned Law determines in Art. 6 *"The bodies and entities of the Union, the States, the Federal District, the Territories and the Municipalities, as well as the foundations instituted by the Public Power, responsible for the protection and improvement of the environmental quality, will constitute the National Environment System - SISNAMA, structured like this:*

"IV - executing agencies: the Brazilian Institute for the Environment and Renewable Natural Resources - IBAMA and the Chico Mendes Institute for Biodiversity Conservation - Chico Mendes Institute, with the purpose of executing and enforcing the governmental policy and guidelines established for the environment, according to their respective competences; "

Furthermore, the Law provides in Art. 9 that *"The following are instruments of the National Environmental Policy:*

VI - the creation of territorial spaces specially protected by the federal, state and municipal government, such as areas of environmental protection, of relevant ecological interest and extractive reserves."

- **Forest Code, Law 12,651 of 2012.** Provides for the protection of native vegetation; In Art. 12 that:

“Every rural property must maintain an area with native vegetation cover, as a Legal Reserve, without prejudice to the application of the rules on Permanent Preservation Areas, observing the following minimum percentages in relation to the area of the property, except for the cases provided for in art. 68 of this Law:

I - located in the Legal Amazon:

- a) 80% (eighty percent), in the property located in a forested area;*
- b) 35% (thirty-five percent), in the property located in a cerrado area;*
- c) 20% (twenty percent), in the property located in an area of general fields;*

Also, article 4 provides that *"A Permanent Preservation Area is considered, in rural or urban areas, for the purposes of this Law:*

I - the marginal strips of any perennial and intermittent natural watercourse, excluding ephemeral ones, from the edge of the channel of the regular bed, with a minimum width of:

- a) 30 (thirty) meters, for water courses less than 10 (ten) meters wide;*
- b) 50 (fifty) meters, for water courses that are from 10 (ten) to 50 (fifty) meters wide;*
- c) 100 (one hundred) meters, for water courses that are 50 (fifty) to 200 (two hundred) meters wide;*
- d) 200 (two hundred) meters, for water courses that are from 200 (two hundred) to 600 (six hundred) meters wide;*
- e) 500 (five hundred) meters, for water courses that are more than 600 (six hundred) meters wide;*

II - the areas surrounding natural lakes and ponds, in a strip with a minimum width of:

- a) 100 (one hundred) meters, in rural areas, except for water bodies with up to 20 (twenty) hectares of surface, whose marginal range will be 50 (fifty) meters;"*

- **National Policy on Climate Change, provided for in Law 12,187** of December 29, 2009.
- **National Payment Policy for Environmental Services, provided for in Law 14,119** of January 13, 2021. This Law establishes the National Payment Registry for Environmental Services (CNPSA) and the Federal Payment for Environmental Services Program (PFPSA).
- **Civil Code, instituted by Law 10,406** of January 10, 2002. Article 1245 *“Transfers between living persons to property Civil Code, instituted by Law 10,406 of January 10, 2002.”.*

Art. 1,238. Whoever, for fifteen years, without interruption or opposition, owns a property as his own, acquires the property, regardless of title and good faith; being able to request the judge to declare it so by sentence, which will serve as a title for the registration in the Real Estate Registry Office.

Art. 1,242. The person who, continuously and undisputedly, with just title and in good faith, owns it for ten years also acquires the property.

- **Civil Procedure Code, Law 13.105** of March 16, 2015.
- **Statute of the Indigenous, Law 6001** of December 19, 1973.
- **MMA Ordinance No. 148**, of June 7, 2022. **National List of Endangered Species.**
- **National System of Nature Conservation Units**, regulated in Law No. 9,985 of July 18, 2000.
- **National System of Nature Conservation Units**, regulated in Decree No. 5,746, of April 5, 2006.
- **Biodiversity Law, provided for in Law No. 13,123**, of May 20, 2015. Provides for access to genetic heritage, protection and access to associated traditional knowledge and benefit sharing for the conservation and sustainable use of biodiversity.
- **Ordinance No. 51, of July 15, 2022.** Topic: Opening of trails.
- **National Policy for Sustainable Development of Traditional Peoples and Communities**, instituted by Decree 6040, of February 7, 2007. Art. 3. *“I - Traditional Peoples and Communities: culturally differentiated groups that are recognized as such, that have their own forms of social organization, that occupy and use territories and natural resources as a condition for their cultural, social, religious, ancestral and economic reproduction, using knowledge, innovations and practices generated and transmitted by tradition;”*

III - Sustainable Development: the balanced use of natural resources, aimed at improving the quality of life of the present generation, guaranteeing the same possibilities for future generations.

- **Consolidation of Labor Laws (CLT), Decree-Law No. 5,452**, May 1, 1943.
- **Regulatory Norms for Rural Work**, established by Law 5,889 of June 8, 1973.

Amazon State Laws

- **State Law No. 3,785**, July 24, 2012, provides for environmental licensing in the state of Amazonas.
- **State Law No. 3,789**, July 27, 2012, provides for forest replacement in the state of Amazonas.
- **State Decree No. 32,986**, November 30, 2012. Regulates Law No. 3,789/2012, which provides for forest replacement in the state of Amazonas.

- **State Law No. 4,406**, December 13, 2016. Establishes the State Environmental Regularization Policy and provides for the Rural Environmental Registry (*Cadastro Ambiental Rural* - CAR), the Rural Environmental Registry System (*Sistema de Cadastro Ambiental Rural* - SICAR-AM) and the Environmental Regularization Program (*Programa de Regularização Ambiental* - PRA), in the state of Amazonas.

Compliance

Due to the project category being REDD+ and the area conservation contract signed by the landowner, the project declares that it observes the guidelines established regarding the Conservation Unit (UC) of the Private Natural Heritage Reserve (RPPN) type, as specified in Law No. 9985/2000 and Decree No. 5746/2006.

The activities implemented in the project area are considered to be of a light, irrelevant or non-existent risk level, with no forecast of causing significant impacts and which are independent of an authorizing act in the environmental licensing process, which does not exempt the stakeholder from the requirements eventually required by other entities.

The activities of opening trails in the vegetation followed what was indicated by Ordinance No. 51, of July 15, 2022, edited by the Ministry of the Environment/Brazilian Institute of the Environment and Renewable Natural Resources.

The Chico Mendes Institute for Biodiversity Conservation (ICMBio) is an executing body that is part of the National Environment System (Sisnama). As explained by ICMBio: "When the collection of botanical, fungal or microbiological material is carried out outside a federal conservation unit or does not involve endangered species, the researcher can request proof of 'Voluntary Registration' through Sisbio."

Brazilian legislation does not provide for authorization (or license) for the collection and transport of botanical, fungal and microbiological material for scientific or educational purposes, when these are not carried out within a federal conservation unit or do not involve endangered species. However, voluntary registration was proposed through SISBio. The Açaí project area does not overlap with any conservation unit and the collection of sample material does not require authorization.

When necessary for the purpose of botanical identification of tree species, specimens containing branches with leaves, flowers or fruits were collected. The responsible researcher has active licenses for the collection of botanical samples in the state of Amazonas issued by SISBio/ICMBio and SISGEN.

We declare that the project proponents do not have any issues that would hinder their execution or pose risks to the project. Both entities hold a negative embargo certificate issued by IBAMA.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

☐ Yes ☒ No

1.16.3 Projects Rejected by Other GHG Programs

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

☐ Yes ☒ No

1.17.2 No Double Claiming with Other Forms of Environmental Credit

☐ Yes ☒ No

1.17.3 Supply Chain (Scope 3) Emissions

☐ Yes ☒ No

1.18 Sustainable Development Contributions

The Açaí REDD+ project aims to promote sustainable development in the region, driven by collaborative efforts among all stakeholders. The project will contribute to the United Nations Sustainable Development Goals (SDGs) described below.

	<u>Goal 2. End hunger, achieve food security and improved nutrition and promote sustainable agriculture</u> The project enables actions to strengthen collective participatory management and community capacity-building in association with different partners and extensionists, aiming to promote knowledge about the importance of reconciling good production practices with the preservation and maintenance of natural resources. Monitoring indicators will be established for the actions and activities carried out throughout the project, as well as their evidence through photographic records and attendance lists.
	<u>Goal 4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all</u> The project facilitates and encourages access to education through training courses focused on the environmental and socioeconomic areas, especially on practices of sustainable extraction of non-timber forest products, associative financial and administrative education and entrepreneurship. Indicators for monitoring the actions and activities developed throughout the project will be established, as well as their evidence through photographic records and attendance list.
	<u>Goal 12. Ensure sustainable consumption and production patterns</u> The project aims to promote sustainable practices through actions aimed at extractivism and low-impact non-timber forest management, according to the demand and profile of local communities. In this sense, among the actions that will be implemented by the project are the dissemination of knowledge, instructions and experiences focused on the efficient and sustainable use of natural resources⁴¹ and environmental preservation.
	<u>Goal 13. Take urgent action to combat climate change and its impacts</u> The project aims to avoid the emission of greenhouse gases from deforestation and forest degradation through forest conservation activities, such as: monitoring of local biodiversity through forest inventories, monitoring of forest cover and surveillance of the area through satellite images, and surveillance of the area via river patrol along rivers and creeks.

⁴¹ Brazilian Indicators for the Sustainable Development Goals. SDG 12. Available at: <https://odsbrasil.gov.br/objetivo/objetivo?n=12>. Accessed on: 05/07/2023

	<u>Goal 15. Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss</u> The project aims to conserve the Amazon rainforest and biodiversity by implementing inspection measures in an area known for high deforestation rates and ecological significance. Conservation efforts involve monitoring biodiversity, forest cover, and conducting surveillance using satellite imagery and river patrols.
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Brazilian NDCs⁴² aim to reduce emissions by 48% by 2025 and 53% by 2030 compared to 2005, with the additional goal of achieving net-zero emissions by 2050. The Açaí REDD+ project plays a crucial role in addressing challenges such as land regularization, combating illegal deforestation, and contributing to international targets.

1.19 Additional Information Relevant to the Project

1.19.1 Leakage Management

Section not required for under development.

1.19.2 Commercially Sensitive Information

Section not required for under development.

1.19.3 Further Information

Section not required for under development.

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Section not required for under development.

**Stakeholder
Identification**

⁴² Nationally Determined Contributions.

Legal or customary tenure/access rights	
Stakeholder diversity and changes over time	
Expected changes in well-being	
Location of stakeholders	
Location of resources	

2.1.2 Stakeholder Consultation and Ongoing Communication

Section not required for under development.

Date of stakeholder consultation	
Stakeholder engagement process	
Consultation outcome	
Ongoing communication	
Stakeholder input	

2.1.3 Free Prior and Informed Consent

Section not required for under development.

Obtaining consent	
Outcome of FPIC	

2.1.4 Grievance Redress Procedure

Section not required for under development.

Development process	
Grievance redress procedure	

2.1.5 Public Comments

Section not required for under development.

Comments received	Actions taken

2.2 Risks to Stakeholders and the Environment

Section not required for under development.

	Risks identified	Mitigation or preventative measure taken
Risks to stakeholder participation		
Working conditions		
Safety of women and girls		
Safety of minority and marginalized groups, including children		
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials)		

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

Section not required for under development.

Discrimination and sexual harassment	
Management experience	
Gender equity in labor and work	
Human trafficking, forced labor, and child labor	

2.3.2 Human Rights

Section not required for under development.

2.3.3 Indigenous Peoples and Cultural Heritage

Section not required for under development.

2.3.4 Property Rights

Section not required for under development.

Rights to territories and resources	
Respect for property rights	

2.3.5 Benefit Sharing

Section not required for under development.

Process used to design the benefit sharing plan	
Summary of the benefit sharing plan	

Approval and dissemination of benefit sharing plan

2.4 Ecosystem Health

Section not required for under development.

	Risks identified	Mitigation or preventative measure taken
Impacts on biodiversity and ecosystems		
Soil degradation and soil erosion		
Water consumption and stress		
Usage of fertilizers		

2.4.1 Rare, Threatened, and Endangered species

Section not required for under development.

Species and habitat	
...	

2.4.2 Introduction of species

Section not required for under development.

Species introduced	Classification	Justification for use	Adverse effects and mitigation

Existing invasive species	Mitigation measures to prevent spread or continued existence of invasive species

2.4.3 Ecosystem conversion

Section not required for under development.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Table 3.1. Methodology, tools and modules applied to the Açaí REDD+ project.

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Methodology	VM0007	VM0007 REDD+ Methodology Framework (REDD+MF),	1.7
Module (carbon pool)	VMD0001	Estimation of carbon stocks in the above and belowground biomass in live tree and non tree pools (CP AB)	1.2
Module (carbon pool)	VMD0005	Estimation of carbon stocks in the long term wood products pool (CP W)	1.1
Module (baseline)	VMD0007	Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation (BL UP)	3.3
Module (leakage)	VMD0010	Estimation of emissions from activity shifting for avoiding unplanned deforestation and avoiding unplanned wetland degradation (LK ASU)	1.2
Module (leakage)	VMD0011	Estimation of emissions from market effects (LK ME)	1.2
Module (emissions)	VMD0013	Estimation of greenhouse gas emissions from biomass and peat burning (E BPB)	1.3
Module (monitoring)	VMD0015	Methods for monitoring of greenhouse gas emissions and removals in REDD project activities (M-REDD)	2.3
Module (miscellaneous)	VMD0016	Methods for stratification of the project area (X STR)	1.3

Module (miscellaneous)	VMD0017	Estimation of uncertainty for REDD project activities (X UNC)	2.2
Tool	T-ADD	Combined tool to identify the baseline scenario and demonstrate additionality in A/R CDM project activities	1.0
Tool	T-SIG	Tool for testing significance of GHG emissions in A/R CDM project activities	1.0
Tool	T-BAR	AFOLU Non-Permanence Risk Tool	4.2

3.2 Applicability of Methodology

Table 3.2. Applicability criteria for the Açaí REDD+ project.

Reference ID/Title	Applicability condition	Justification of conformance
VM0007	This REDD+ activity is applicable due to the following conditions: The land in the project area has been classified as forest for at least 10 years before the project start date; Where land within the project area is peatland or tidal wetlands and emissions from the SOC pool are deemed significant, the relevant WRC modules are applied alongside other relevant modules. The baseline deforestation in the project area falls under the category of Unplanned Deforestation (VCS AUD category) Leakage prevention activities do not include: i) Agricultural lands that are flooded to increase production (e.g., rice paddies); ii) Intensification of livestock production through the use of feedlots and/or manure lagoons.	The lands in the Açaí REDD+ project area have been qualified as forests for at least 10 years prior to the project start date. This classification adheres to the forest definition outlined by FAO/UNFCCC⁴³ and the specific thresholds for the Brazilian Amazon: a minimum area of 0.5 hectares, a minimum height of 5 meters, and a minimum canopy cover of 10%. The baseline deforestation type in the project area falls under the category of Unplanned Deforestation (VCS AUD category)

⁴³ FAO - Forests and Climate Change Working Paper 17. Available at: <<https://www.fao.org/3/CA0176EN/ca0176en.pdf>>. Accessed on: 13/10/2023

VMD0001	The module allows for ex-ante estimation of carbon stocks in aboveground and belowground tree and non-tree biomass in the baseline scenario (both pre and post-deforestation stocks) as well as in the project case, and for ex-post estimation of changes in carbon stocks of aboveground and belowground tree biomass in the project case. The module applies to all forest types and age classes. The inclusion of the aboveground tree biomass reservoir as part of the project is mandatory. Aboveground non-tree biomass must be included as part of the project boundary if the following applicability criteria are met: • Aboveground non-tree biomass stocks are higher in the baseline than in the project scenario, and • Aboveground non-tree biomass is considered significant (using the T-SIG module).	This module applies to all forest types and age classes. The inclusion of the aboveground tree biomass reservoir as part of the project boundaries is mandatory according to the REDD-MF module.
VMD0005	This module allows for ex-ante estimation of carbon stocks in the long-term wood products pool in the baseline scenario. The carbon stocks addressed here are those remaining in wood products after 100 years; most emissions associated with harvesting, processing, and wood waste, as well as eventual product removal, occur within this timeframe, and this module employs the simplifying assumption that the remaining proportion after 100 years is effectively 'permanent'.	This module is applicable to all forest types and age classes. Inclusion of the aboveground tree biomass pool as part of the project boundary is mandatory as per the REDD-MF framework module.
VMD0007	This module allows estimating changes in carbon stocks and GHG emissions related to unplanned deforestation and unplanned wetland degradation in the baseline scenario. Forest degradation is not considered.	• Their practices are predominantly illegal and not within their legal rights

	The following conditions must be met to apply this module. The forest landscape configuration can be mosaic, transition, or frontier. The module must be applied to all project activities where baseline deforestation agents: (i) Clear land for settlements, agricultural production (farmers), livestock farming, or aquaculture, where such deforestation for agricultural, livestock farming, or aquaculture purposes does not represent large-scale industrial agriculture or aquaculture activities. (ii) Have no legally documented and undisputed right to deforest the land for these purposes. (iii) Are residents in the region (reference region) or immigrants. If, in the baseline scenario to prevent unplanned deforestation project activities, post-deforestation land use constitutes reforestation, this methodology cannot be used. Where, before the project, unsustainable firewood collection is occurring within the project boundaries, modules BL-DFW and LK-DFW must be used to determine potential leakage.	
VMD0010	This module provides methods for estimating emissions from displacement of unplanned deforestation and degradation of wetlands (leakage due to activity change). Activities subject to potential displacement include the conversion of forested areas to pastures, crops, and other land uses, or the conversion of intact or partially degraded wetlands to drained or degraded wetlands. The module is mandatory if Module BL-UP is used to define the baseline,	The Açaí REDD+ project has an area dedicated to avoiding unplanned deforestation, using the BL-UP approach, thus requiring the mandatory application of Module LK-ASU.

	and the applicability conditions in Module BL-UP must be fully met.	
VMD0011	The module allows estimating GHG emissions caused by market effects leakage related to the logging of sawn wood, firewood, or charcoal in the baseline for carbon projects. It's applicable to calculate the market effects leakage of REDD projects that must substantially and permanently reduce wood extraction levels. When REDD project activities result in reductions in timber harvesting, it's likely that production may be displaced to other areas of the country to compensate for the reduction. As referenced in the REDD-MF Framework, the module is mandatory where: • The deforestation process involves timber harvesting for commercial markets. • The baseline is calculated using BL-DFW AND firewood or charcoal is harvested for commercial markets. In all other circumstances, the module should not be used.	The module is mandatory when the deforestation process involves wood extraction for commercial markets.
VMD0013	This module provides a step-by-step approach to estimate greenhouse gas emissions from biomass and peat burning. This module is applicable to REDD project activities with biomass burning emissions and REDD-WRC project activities with emissions from biomass and/or peat burning. This module is also applicable to RWE and ARR-RWE project activities with emissions from peat burning.	In the baseline scenario, fire is used to clear the land, resulting in emissions of CO₂, N₂O and CH₄. When used in the baseline, accounting should occur both under the baseline and in the project scenario and both within the project area and in the leakage belt. Where post-fire incidents occur in areas that coincide with deforested or degraded areas in the baseline, the module should be used to account for greenhouse gas emissions.
VMD0016	This module provides guidance on how to stratify the project area into discrete and relatively homogeneous units to enhance the accuracy and precision of	Strata are only used for forest classes under deforestation pressure and are the same for both the baseline and project scenarios.

	carbon stock and carbon stock change estimates.	
VMD0017	The module allows estimating uncertainty in baseline estimates and project sequestration, emissions, and leaks estimates. The module focuses on the following sources of uncertainty: • Determination of deforestation and degradation rates. • Uncertainty associated with carbon stock estimates and changes in carbon reservoirs. • Uncertainty associated with peat emissions estimation. • Uncertainty in project emissions assessment. When an uncertainty value is unknown or cannot be readily calculated, the project must justify the use of a decidedly conservative value, and an uncertainty of 0% can be applied to this component.	This module is mandatory when using the REDD+ MF methodology.
T-ADD	AFOLU activities identical or similar to the proposed project activity on the site within the proposed project boundary, whether executed with or without registration as a VCS AFOLU project, will not result in the violation of any applicable laws, even if the law is not enforced. The use of this tool to determine additionality requires the baseline methodology to provide a gradual approach that justifies the determination of the most plausible baseline scenario. Project proponents proposing new baseline methodologies must ensure consistency between determining a baseline scenario and determining the additionality of a project activity.	As stated in section 2.3.16 , the project is in compliance with all relevant laws, statutes, and regulatory frameworks.

3.3 Project Boundary

Section not required for under development.

Source		Gas	Included?	Justification/Explanation
Baseline	Source 1	CO ₂		
		CH ₄		
		N ₂ O		
		Other		
	Source 2	CO ₂		
		CH ₄		
		N ₂ O		
		Other		
Project	Source 1	CO ₂		
		CH ₄		
		N ₂ O		
		Other		
	Source 2	CO ₂		
		CH ₄		
		N ₂ O		
		Other		

3.4 Baseline Scenario

Section not required for under development.

3.5 Additionality

Section not required for under development.

3.5.1 Regulatory Surplus

Section not required for under development.

3.5.2 Additionality Methods

Section not required for under development.

3.6 Methodology Deviations

Section not required for under development.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Section not required for under development.

4.2 Project Emissions

Section not required for under development.

4.3 Leakage Emissions

Section not required for under development.

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

Section not required for under development.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)

State the non-permanence risk rating (%)

Example: 20%

Has the non-permanence risk report been attached as either an appendix or a separate document?	☐ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e, the Long-term Average (LTA).	
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☐ No If no, provide justification.
State, in tCO ₂ e, the expected total GHG benefit to date.	
Is the number of GHG credits issued below the LTA?	☐ Yes ☐ No If no, provide justification.

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated buffer pool allocation (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCU issuance (tCO ₂ e)
							Example: 16,000

5 MONITORING

5.1 Data and Parameters Available at Validation

Section not required for under development.

Data / Parameter	
Data unit	
Description	

Source of data	
Value applied	
Justification of choice of data or description of measurement methods and procedures applied	
Purpose of data	
Comments	

5.2 Data and Parameters Monitored

Section not required for under development.

Data / Parameter	
Data unit	
Description	
Source of data	
Description of measurement methods and procedures to be applied	
Frequency of monitoring/recording	
Value applied	
Monitoring equipment	
QA/QC procedures to be applied	
Purpose of data	
Calculation method	
Comments	

5.3 Monitoring Plan

Section not required for under development.

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Section not required for under development.

Section	Information	Justification

